neel.kp305@gmail.com
+91 9408688946

Neel Patel

LinkedIn
Website

Education

Institute of Technology, Nirma University — Ahmedabad, India
B.Tech in Electronics and Communication Engineering, CGPA: 9.40/10.00 (Dept. Rank: 2) — *Aug 2023 – May 2027*

- **Minor:** Computer Science and Engineering (CSE)
- **Coursework:** Computer Architecture, Operating Systems, Machine Learning, Digital Logic Design, VLSI, FPGA-based System Design, Computer Networks

Skills

Programming: C++, C, Python, MATLAB, Shell Scripting, Assembly
GPU & Computer Architecture: GPU Memory Systems, CPU Architecture, Parallel Computing, Memory Systems, TLBs, Cache Hierarchies, ARM ISA
AI & Accelerator Systems: Systolic Arrays, Transformer Architectures, Attention Mechanisms, KV-Cache
HDL & Digital Design: Verilog, SystemVerilog, RTL Design, CDC, RTL Linting
Architecture & EDA Tools: gem5, MGPUSim, Cadence Virtuoso, Genus, Innovus, Synopsys VCS, Verdi, Spyglass, Siemens Tessent, Vivado, Quartus

Research Experience

Durham University, UK — Remote
Research Intern – Computer Architecture & GPU Memory Systems — *Jun 2026 – Present*

- Investigating STT-RAM-based GPU L2 TLB architectures for efficient address translation and memory management, targeting AMD MI300-series GPU architectures.
- Evaluating GPU TLB organizations and memory-hierarchy trade-offs across performance, power, and area to identify optimization opportunities.
- Conducting cycle-accurate simulations using gem5 and MGPUSim to characterize GPU TLB behavior and identify memory-system bottlenecks.
- Designing and routing multi-level GPU TLB interconnects (L1/L2/L3), modeling asymmetric STT-RAM read/write latencies and evaluating architectural optimizations.

University of Arizona, USA — Remote
Transformer & KV-Cache Optimization Study — *2026*

- Conducted a detailed literature review of Transformer architectures, attention mechanisms, and LLM inference optimization, focusing on memory and data-movement bottlenecks.
- Studied KV-cache optimization and memory-efficient attention techniques to understand their role in reducing memory overhead during LLM inference.

MeitY / Chips to Startup (C2S) Research Project
Secure and Energy-Efficient Mixed-Domain Compute-in-Memory-Based AI Accelerator Chip for Edge Applications

- Contributing to SRAM circuit design and memory-system research for energy-efficient compute-in-memory-based AI acceleration
- Studying memory design considerations for hardware-efficient edge AI workloads.

Projects

16×16 Weight-Stationary GEMM Systolic Array:

- Designed and implemented a 16×16 weight-stationary systolic array with 256 processing elements in Verilog using Vivado for parallel matrix multiplication, exploiting spatial parallelism and data reuse for AI workloads.
- Developed and verified the RTL datapath and control logic, analyzing pipeline latency, throughput, and processing-element utilization to characterize accelerator performance.
- Achieved 52.3 GOPS at 300 MHz in cycle-accurate simulation, evaluating compute throughput and memory/data-movement trade-offs for matrix workloads.

Mini Operating System Simulator:

- Developed a modular C++ OS simulator implementing PCB-based process management, process-state transitions, I/O blocking, and Round Robin, Priority, and SJF CPU scheduling.
- Implemented virtual memory, two-level paging, virtual-to-physical address translation, page replacement, memory protection, and mutex-based synchronization.

C++ Expression Compiler:

- Developed a modular C++ expression compiler supporting lexical analysis, parsing, expression evaluation, and intermediate representation generation.
- Implemented AST-based expression processing and symbol-table management, translating parsed expressions into an intermediate representation for subsequent processing.

Asynchronous FIFO - RTL Design & Verification:

- Designed a parameterized asynchronous FIFO in Verilog using 2-stage synchronizers and Gray-code pointers for clock-domain crossing.
- Developed SystemVerilog testbenches and verified functionality using Synopsys VCS/Verdi, with SpyGlass CDC analysis and RTL linting.

4×4 SRAM Design Using 6T and 8T Cells:

- Designed a 4×4 SRAM array using 6T and 8T cells in 90 nm technology using Cadence Virtuoso, including sense amplifiers, precharge circuits, write drivers, and decoders.
- Performed SRAM performance and simulation analysis, evaluating the implemented memory architecture and peripheral circuitry.

Industry Experience

eInfochips Ahmedabad, India
Hardware Intern *May 2026 – Jun 2026*

- Executed 90 nm ASIC RTL-to-GDSII flow using Verilog, Cadence NCLaunch, Genus, and Innovus; designed and verified 32-bit RCA and Universal Shift Register.

Ministry of Electronics and Information Technology (MeitY) Ahmedabad, India
DFT Intern *May 2025 – Jun 2025*

- Implemented MBIST using Siemens Tessent and analyzed stuck-at, retention, coupling, and enable faults in QuestaSim; developed LBIST patterns using Cadence Modus and performed Conformal LEC.

Publications

Journal Publications

- **Schmitt Trigger-Based SRAM Architectures (Submitted to IEEE Transactions on Very Large Scale Integration Systems (TVLSI))**: Designed and implemented a 12T Schmitt Trigger-based SRAM in 45 nm technology using Cadence Virtuoso, achieving 1.9× higher stability (RSNM and HSNM) than 8T SRAM, with read/write speed and power performance comparable to recent SRAM design advancements.

Conference Publications

- **Deep Learning-based Framework for Detecting Malicious Proof-of-Stake Blocks in Gaming Transactions**: Developed a deep learning framework to detect and prevent malicious Proof-of-Stake blocks in gaming transaction systems. *IEEE 10th International Conference on Smart and Sustainable Technologies (SpliTech), Bol and Split, Croatia, 2025.*
- **SeeD: AI-based Secure Data Exchange Framework for Connected Electric Vehicles**: Presented a framework enhancing SDN security for Connected Electric Vehicles using AI models for malicious data detection. *IEEE 12th Region 10 Humanitarian Technology Conference (R10-HTC), Kuala Lumpur, Malaysia, 2024.*

Achievements

Certificate of Excellence (Top 1% - Department Rank: 2/220): Secured 2nd highest academic standing in the ECE department across Semesters I–VI.

Certifications & Courses

Digital Design and Verification Academic Curriculum v1.0: Cadence
Basic Static Timing Analysis v3.0: Cadence

Positions of Responsibility

Technical Head, WIE IEEE SBNU: Organized technical events, workshops, and student activities.
Volunteer, Asian Test Symposium (ATS) 2024: Supported conference coordination and event execution involving researchers and industry professionals.